

October 28, 2014

MDEQ, WRD-PSM, Grand Rapids
Attn: Ms. Chris Little
350 Ottawa, NW, Unit 10
Grand Rapids, MI 48909

Project: Wastewater Surveys 2014

Dear Ms. Chris Little,

Enclosed is a copy of the laboratory report for the following work order(s) received by TriMatrix Laboratories:

Work Order	Received	Description
1410331	10/15/2014	MI0020257/Cadillac WWTP

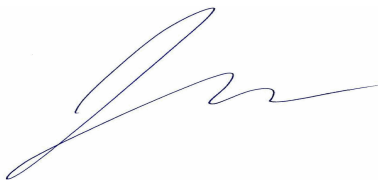
This report relates only to the sample(s) as received. Test results are in compliance with the requirements of the National Environmental Laboratory Accreditation Program (NELAP) and/or one of the following certification programs:

ACCLASS DoD-ELAP/ISO17025 (#ADE-1542); Arkansas DEP (#88-0730/13-049-0); Florida DEP (#E87622-24); Georgia EPD (#E87622-24); Illinois DEP (#200026/003329); Kansas DPH (#E-10302); Kentucky DEP (#0021); Louisiana DEP (#103068); Michigan DPH (#0034); Minnesota DPH (#491715); New York ELAP (#11776/48855); North Carolina DNRE (#659); Texas CEQ (#T104704495-14-4); Virginia DCLS (#460153/2592); Wisconsin DNR (#999472650); USDA Soil Import Permit (#P330-12-00236).

Any qualification or narration of results, including sample acceptance requirements and test exceptions to the above referenced programs, is presented in the Statement of Data Qualifications and Project Technical Narrative sections of this report. Estimates of analytical uncertainties and certification documents for the test results contained within this report are available upon request.

If you have any questions or require further information, please do not hesitate to contact me.

Sincerely,



James D. McFadden
Project Chemist

PROJECT TECHNICAL NARRATIVE(s)**Physical/Chemical Parameters by EPA/APHA/ASTM Methods**

Narrative: Due to sample volumes, batch matrix quality control (QC) was not performed for this analysis. A Method Blank and Laboratory Control Sample comprise the batch QC.

Analysis: SM 5210 B-2011

Sample/Analyte: 1410331-01 CD-001A
1410331-02 CD-001A-F

BOD, Carbonaceous (5-Day)
BOD, Carbonaceous (5-Day)

STATEMENT OF DATA QUALIFICATIONS**Physical/Chemical Parameters by EPA/APHA/ASTM Methods**

Qualification: The nutrient BLK for this CBOD batch was 0.39 mg/L.

Analysis: SM 5210 B-2011

Sample/Analyte: 1411466-BLK1

BOD, Carbonaceous (5-Day)

ANALYTICAL REPORT

Client: **MDEQ, WRD-PSM, Grand Rapids**
 Project: Wastewater Surveys 2014
 Client Sample ID: **CD-001A**
 Lab Sample ID: **1410331-01**
 Matrix: Waste Water

Work Order: **1410331**
 Description: MI0020257/Cadillac WWTP
 Sampled: 10/15/14 10:10
 Sampled By: C.Little
 Received: 10/15/14 15:40

Physical/Chemical Parameters by EPA/APHA/ASTM Methods

Analyte	Analytical Result	RL	MDL	Unit	Dilution Factor	Method	Date Time Analyzed	By	QC Batch
BOD, Carbonaceous (5-Day)	4.8	2.0	1.0	mg/L	1	SM 5210 B-2011	10/16/14 17:19	SKA	1411466

ANALYTICAL REPORT

Client: **MDEQ, WRD-PSM, Grand Rapids**
 Project: Wastewater Surveys 2014
 Client Sample ID: **CD-001A-F**
 Lab Sample ID: **1410331-02**
 Matrix: Waste Water

Work Order: **1410331**
 Description: MI0020257/Cadillac WWTP
 Sampled: 10/15/14 07:00
 Sampled By: C.Little
 Received: 10/15/14 15:40

Physical/Chemical Parameters by EPA/APHA/ASTM Methods

Analyte	Analytical Result	RL	MDL	Unit	Dilution Factor	Method	Date Time Analyzed	By	QC Batch
BOD, Carbonaceous (5-Day)	5.5	2.0	1.0	mg/L	1	SM 5210 B-2011	10/16/14 17:31	SKA	1411466

QUALITY CONTROL REPORT

Physical/Chemical Parameters by EPA/APHA/ASTM Methods

QC Type	Sample Conc.	Spike Qty.	Result	Unit	Spike % Rec.	Control Limits	RPD	RPD Limits	RL	MDL
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Analyte: BOD, Carbonaceous (5-Day)/SM 5210 B-2011

QC Batch: 1411466 (General Inorganic Prep)

Analyzed: 10/16/2014 By: SKA

*Method Blank			ND U	mg/L					2.0	1.0
Laboratory Control Sample		198	208	mg/L	105	85-115			2.0	1.0

W0#1410331

TriMatrix 13360

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ENVIRONMENTAL LABORATORY - ANALYSIS REQUEST SHEET

1-10

Lab Work Order Number

Project Name

Matrix

Cadillac WWTP

WASTE WATER

Site Code/Project Number

AY

CC Email 1 to send additional reports to

Project TAT Days

Sample Collector

MI0020257

15

Dept-Division-District

Index

CC Email 2 to send additional reports to

Project Due Date

Sample collector Phone

MDEQ-WRD-PSM-GR

63206

PCA

CC Email 3 to send additional reports to

cell 616-550-7668

State Project Manager

42202
42203

Chris C. Little

State Project Manager Email

481000

littlec@michigan.gov

State Project Manager Phone

cell 616-550-7668

Phase

00

Overflow Lab Choice 1

Accept Analysis
hold time codes: Yes/No

Contract Firm Primary Contact

Primary Contact Phone

Lab Use Only	Field Sample Identification	Collection Date	Collection Time	Container Count	Comments
1	CD-001A	101514	1010		
2	CD-001A-F	101514	0700		
3					
4					
5					
6					
7					
8					
9					
10					

CBOD 5

00 4 5 6 7 8 9 10

Chain of Custody	Relinquished by	Received By	Date / Time
	Print Name & Org.	Signature	
	Signature		10/15/14 1540
	Print Name & Org.	Signature	
	Print Name & Org.	Signature	
	Print Name & Org.	Signature	

SAMPLE RECEIVING / LOG-IN CHECKLIST



Client MDEQ	New / Add To Project Chemist	Work Order # 1410331
Receipt Record Page/Line # 1-10	Sample #	

Recorded by (initials/date) LR 10/15/14	☒ Cooler ☐ Box ☐ Other	Qty Received 1	Thermometer Used ☒ IR Gun (#202) ☐ Digital Thermometer (#54) ☐ Other (#)	☐ See Additional Cooler Information Form
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Cooler #	Time	Cooler #	Time	Cooler #	Time	Cooler #	Time	
Client	1542							
Custody Seals:		Custody Seals:		Custody Seals:		Custody Seals:		
☒ None ☐ Present / Intact ☐ Present / Not Intact		☐ None ☐ Present / Intact ☐ Present / Not Intact		☐ None ☐ Present / Intact ☐ Present / Not Intact		☐ None ☐ Present / Intact ☐ Present / Not Intact		
Coolant Type:		Coolant Type:		Coolant Type:		Coolant Type:		
☐ Loose Ice ☐ Bagged Ice ☒ Blue Ice ☐ None		☐ Loose Ice ☐ Bagged Ice ☐ Blue Ice ☐ None		☐ Loose Ice ☐ Bagged Ice ☐ Blue Ice ☐ None		☐ Loose Ice ☐ Bagged Ice ☐ Blue Ice ☐ None		
Coolant Location:		Coolant Location:		Coolant Location:		Coolant Location:		
☒ Dispersed / ☐ Top / ☐ Middle / ☐ Bottom		☐ Dispersed / ☐ Top / ☐ Middle / ☐ Bottom		☐ Dispersed / ☐ Top / ☐ Middle / ☐ Bottom		☐ Dispersed / ☐ Top / ☐ Middle / ☐ Bottom		
Temp Blank Present: ☐ Yes ☐ No		Temp Blank Present: ☐ Yes ☐ No		Temp Blank Present: ☐ Yes ☐ No		Temp Blank Present: ☐ Yes ☐ No		
If Present, Temperature Blank Location is:		If Present, Temperature Blank Location is:		If Present, Temperature Blank Location is:		If Present, Temperature Blank Location is:		
☐ Representative ☐ Not Representative		☐ Representative ☐ Not Representative		☐ Representative ☐ Not Representative		☐ Representative ☐ Not Representative		
Observed °C	Correction Factor °C	Actual °C	Observed °C	Correction Factor °C	Actual °C	Observed °C	Correction Factor °C	
Temp Blank:			Temp Blank:			Temp Blank:		
Sample 1:	4.7	-	4.7	Sample 1:		Sample 1:		
Sample 2:	4.6	-	4.6	Sample 2:		Sample 2:		
Sample 3:	5.1	-	5.1	Sample 3:		Sample 3:		
3 Sample Average °C: 4.8			3 Sample Average °C:			3 Sample Average °C:		
☐ Cooler ID on COC? ☐ VOC Trip Blank received?			☐ Cooler ID on COC? ☐ VOC Trip Blank received?			☐ Cooler ID on COC? ☐ VOC Trip Blank received?		

If any shaded areas checked, complete Sample Receiving Non-Conformance and/or Inventory Form

Paperwork Received Yes ☒ No ☐ Chain of Custody record(s)? If No, Initiated By _____ Received for Lab Signed/Date/Time? Shipping document? Other _____ COC Information ☐ TriMatrix COC ☒ Other _____ COC ID Numbers:	Check Sample Preservation N/A ☒ Yes ☐ No ☐ ☒ Temperature Blank OR average sample temperature, 26° C? ☐ If either is 26° C, was thermal preservation required? If "Yes", Project Chemist Approval Initials: _____ If "Yes" Completed Non Con Cooler - Cont Inventory Form? Completed Sample Preservation Verification Form? ☒ Samples chemically preserved correctly? If "No", added orange tag? ☒ Received pre-preserved VOC soils? ☐ MeOH ☐ Na ₂ SO ₄			
Check COC for Accuracy Yes ☒ No ☐ ☐ Analysis Requested? ☒ Sample ID matches COC? ☒ Sample Date and Time matches COC? Container type completed on COC? ☒ All container types indicated are received?	Check for Short Hold-Time Prep/Analyses ☐ Bacteriological ☐ Air Bags ☐ EnCores / Methanol Pre-Preserved ☐ Formaldehyde/Aldehyde ☒ Green-tagged containers ☐ Yellow/White-tagged 1 L ambers (SV Prep-Lab)			
Sample Condition Summary N/A ☒ Yes ☐ No ☐ ☒ Broken containers/lids? ☒ Missing or incomplete labels? ☒ Illegible information on labels? ☒ Low volume received? ☒ Inappropriate or non-TriMatrix containers received? ☐ VOC vials / TOX containers have headspace? ☒ Extra sample locations / containers not listed on COC?	Notes AFTER HOURS ONLY: COPIES OF COC TO LAB AREA(S) ☐ NONE RECEIVED ☒ RECEIVED, COCs TO LAB(S)			
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 33%;"> ☐ Trip Blank received Cooler Received (Date/Time) 10/15/14 1540 </td> <td style="width: 33%;"> ☐ Trip Blank not listed on COC Paperwork Delivered (Date/Time) 10/15/14 1545 </td> <td style="width: 33%;"> ☒ Yes ☐ No ≤1 Hour Goal Met? </td> </tr> </table>		☐ Trip Blank received Cooler Received (Date/Time) 10/15/14 1540	☐ Trip Blank not listed on COC Paperwork Delivered (Date/Time) 10/15/14 1545	☒ Yes ☐ No ≤1 Hour Goal Met?
☐ Trip Blank received Cooler Received (Date/Time) 10/15/14 1540	☐ Trip Blank not listed on COC Paperwork Delivered (Date/Time) 10/15/14 1545	☒ Yes ☐ No ≤1 Hour Goal Met?		

Client: MDEQ	Work Order #: 1410331
Receipt Log #: 1-9	Completed By (Initials/Date): 8/210/15/14
Project Chemist: _____	

COC ID #				Adjusted by: _____ Date: _____		DO NOT ADJUST pH FOR THESE CONTAINER TYPES			
Container Type	5 / 23	4	13	3	6	15			
Tag Color	Lt. Blue	Blue	Brown	Green	Red	Red Stripe			
Preservative	NaOH	H ₂ SO ₄	H ₂ SO ₄	None	HNO ₃	HNO ₃			
Expected pH	>12	<2	<2	6-8	<2	<2			
COC Line #1				✓					
COC Line #2				✓					
COC Line #3									
COC Line #4									
COC Line #5									
COC Line #6									
COC Line #7									
COC Line #8									
COC Line #9									
COC Line #10									

pH Strip Reagent #	4051306
☒	
☐	

Aqueous Samples: For each sample and container type, check the box if pH is acceptable. If pH is not acceptable for any sample container, record pH in box, and note on Sample Receiving Checklist and on Sample Receiving Non-Conformance Form. If approved by Project Chemist, add acid or base to the sample to achieve the correct pH. Add up to, but do not exceed 2x the volume initially added at container prep (see table below for initial volumes used). Add orange pH tag to sample container and record information requested. Record adjusted pH on this form. Do not adjust pH for container types 3, 6, and 15.

COC ID #				Adjusted by: _____ Date: _____		DO NOT ADJUST pH FOR THESE CONTAINER TYPES			
Container Type	5 / 23	4	13	3	6	15			
Tag Color	Lt. Blue	Blue	Brown	Green	Red	Red Stripe			
Preservative	NaOH	H ₂ SO ₄	H ₂ SO ₄	None	HNO ₃	HNO ₃			
Expected pH	>12	<2	<2	6-8	<2	<2			
COC Line #1									
COC Line #2									
COC Line #3									
COC Line #4									
COC Line #5									
COC Line #6									
COC Line #7									
COC Line #8									
COC Line #9									
COC Line #10									

Container Size (mL)	Original Vol. of Preservative (mL)
Container Type 5	NaOH
500	2.5
1000	5.0
Container Type 4	H ₂ SO ₄
125	0.5
250	1.0
500	2.0
1000	4.0
Container Type 13	H ₂ SO ₄
500	2.5